

HEAD INJURY

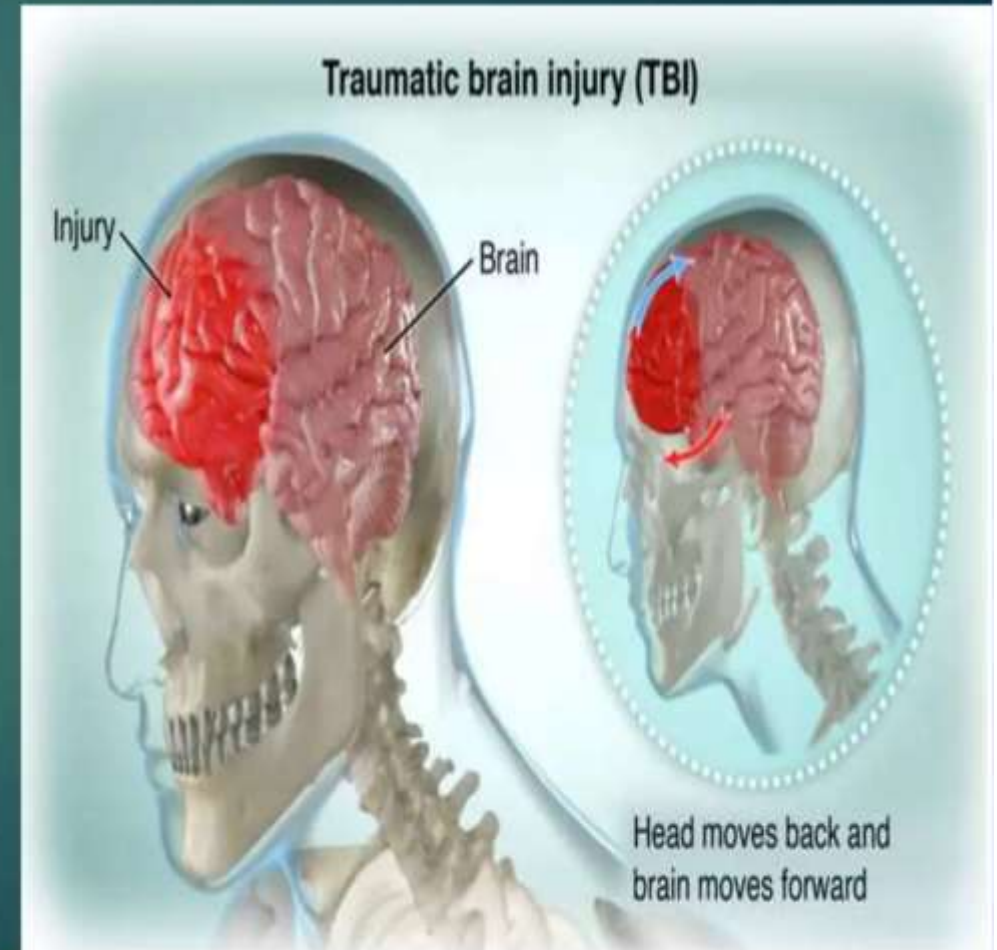


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DESCRIPTION

- ▶ Head injury is trauma to the skull resulting in mild to extensive damage to the brain
- ▶ Immediate complications include cerebral bleeding, hematomas, uncontrolled increased ICP, infections, and seizures
- ▶ Changes in personality or behavior, cranial nerve deficits and any other residual deficits depend on the area of the brain damage.



TYPES OF HEAD INJURIES

OPEN

- ▶ Scalp lacerations
- ▶ Fractures in the skull
- ▶ Interruption of the dura mater

CLOSED

- ▶ Concussions
- ▶ Contusions
- ▶ Fracture

HEAD INJURY - TYPES

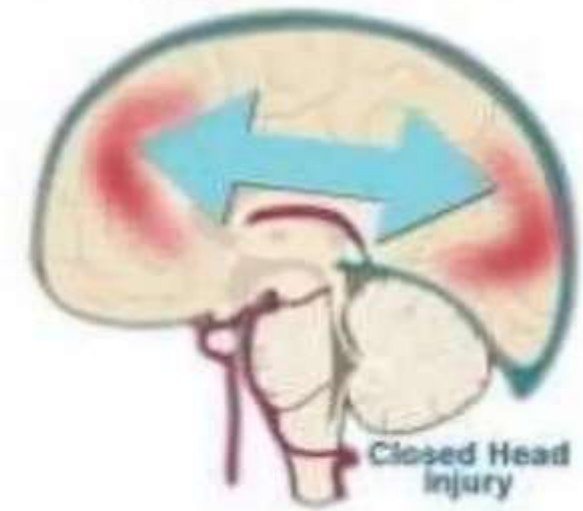
OPEN HEAD INJURY:

There is penetration to the skull.



CLOSED HEAD INJURY

There is NO penetration to the skull.

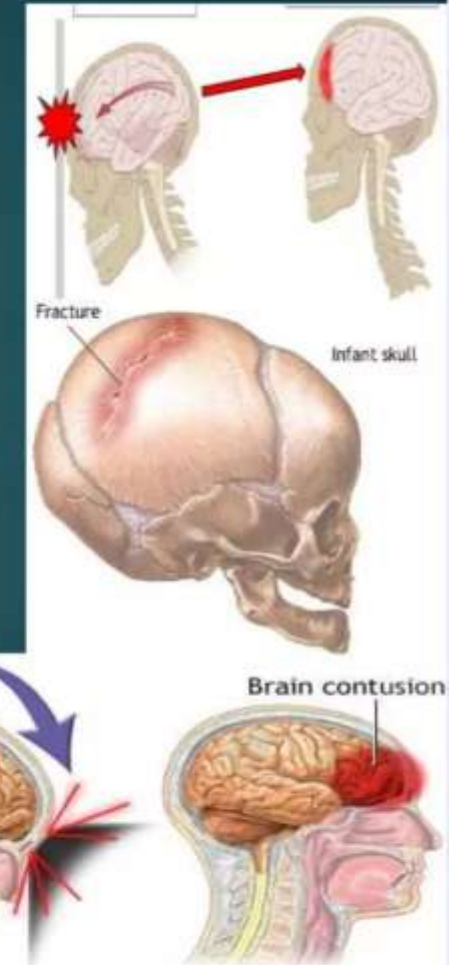


TYPES OF HEAD INJURY

- ▶ **CONCUSSION** :- concussion is jarring of the brain within the skull with temporary loss of consciousness .
- ▶ **CONTUSION** :- Contusion is a bruising type of injury to the brain
Contusion may occur with subdural or extradural collection of blood
- ▶ **SKULL FRACTURES** :- linear, depressed, compound, comminuted
- ▶ **EPIDURAL HEMATOMA** :- The most serious type of hematoma forms rapidly and resulted from arterial bleeding.

Epidural hematoma forms between the dura and the skull from a tear in the meningeal artery .

Epidural hematoma is a surgical emergency.



SUBDURAL HEMATOMA :-subdural hematoma

forms slowly and results from a venous bleed

Subdural hematoma occurs under the dura as

a result of tears in the veins crossing the subdural space

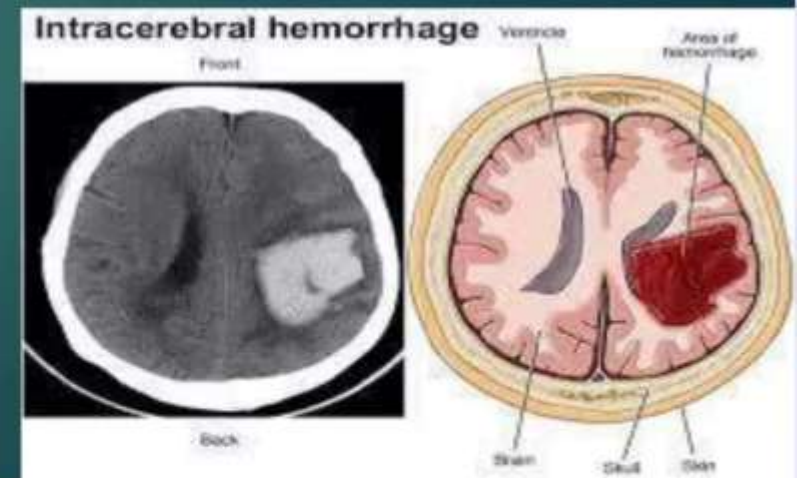
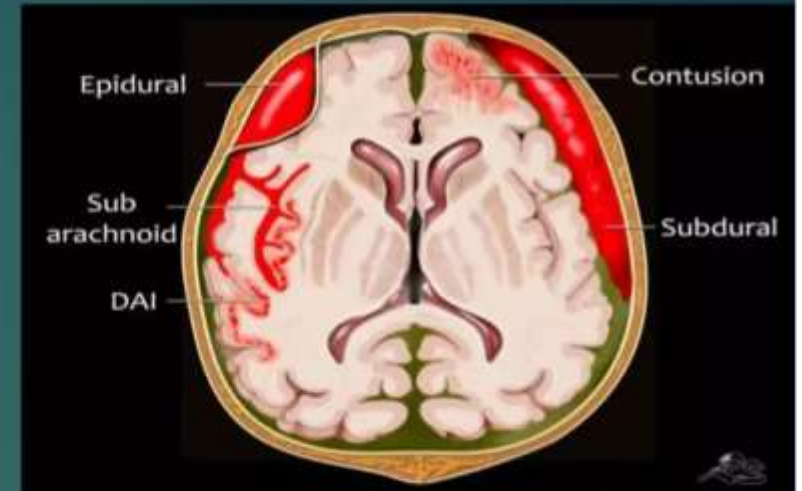
INTRACEREBRAL HEMORRHAGE :- Multiple hemorrhages

occur around a confused area.

SUBARACHNOID HEMORRHAGE :- Bleeding occurs

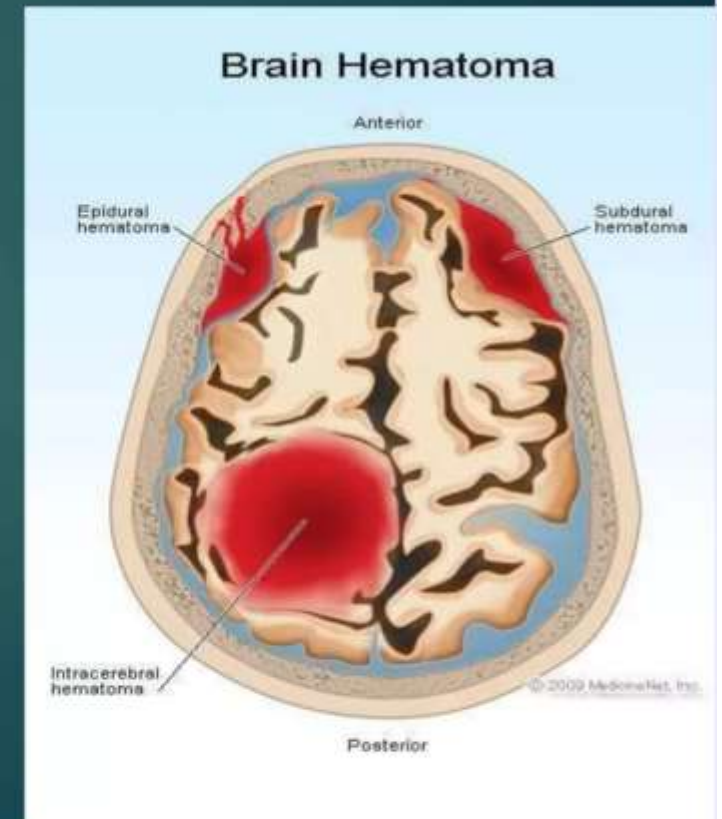
directly into the brain, the ventricles,

or the subarachnoid space.



HEMATOMA

- ▶ **DESCRIPTION** :-- Hematoma can occur as a result of a subarachnoid hemorrhage or an intracerebral hemorrhage
- ▶ **ASSESSMENT** :-
- ▶ Assessment findings depend on the injury
- ▶ Clinical manifestations usually results from increased ICP
- ▶ Changing neurological signs in the client
- ▶ Changes in level of consciousness
- ▶ Airway and breathing pattern changes
- ▶ Vital signs for signs of increasing ICP
- ▶ Headache, nausea, and vomiting



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- ▶ Visual disturbances , pupillary changes , papilledema, and extraocular eye movements
 - ▶ Nuchal rigidity
 - ▶ CSF drainage from the ears or nose
 - ▶ Weakness and paralysis
 - ▶ Posturing
 - ▶ Decreased sensation or absence of feeling
 - ▶ Reflex activity changes
 - ▶ Seizure activity

INTERVENTIONS

- ▶ Monitor respiratory status and maintain a patent airway because increased CO2 levels increase cerebral edema
- ▶ Monitor neurological status and vital signs ,including temperature
- ▶ Monitor for increased ICP
- ▶ Maintain head elevation to reduce venous pressure
- ▶ Prevent neck flexion
- ▶ Initiate normothermia measures for increased temperature
- ▶ Assesses cranial nerve function , reflexes, and motor and sensory function
- ▶ Initiate seizure precautions



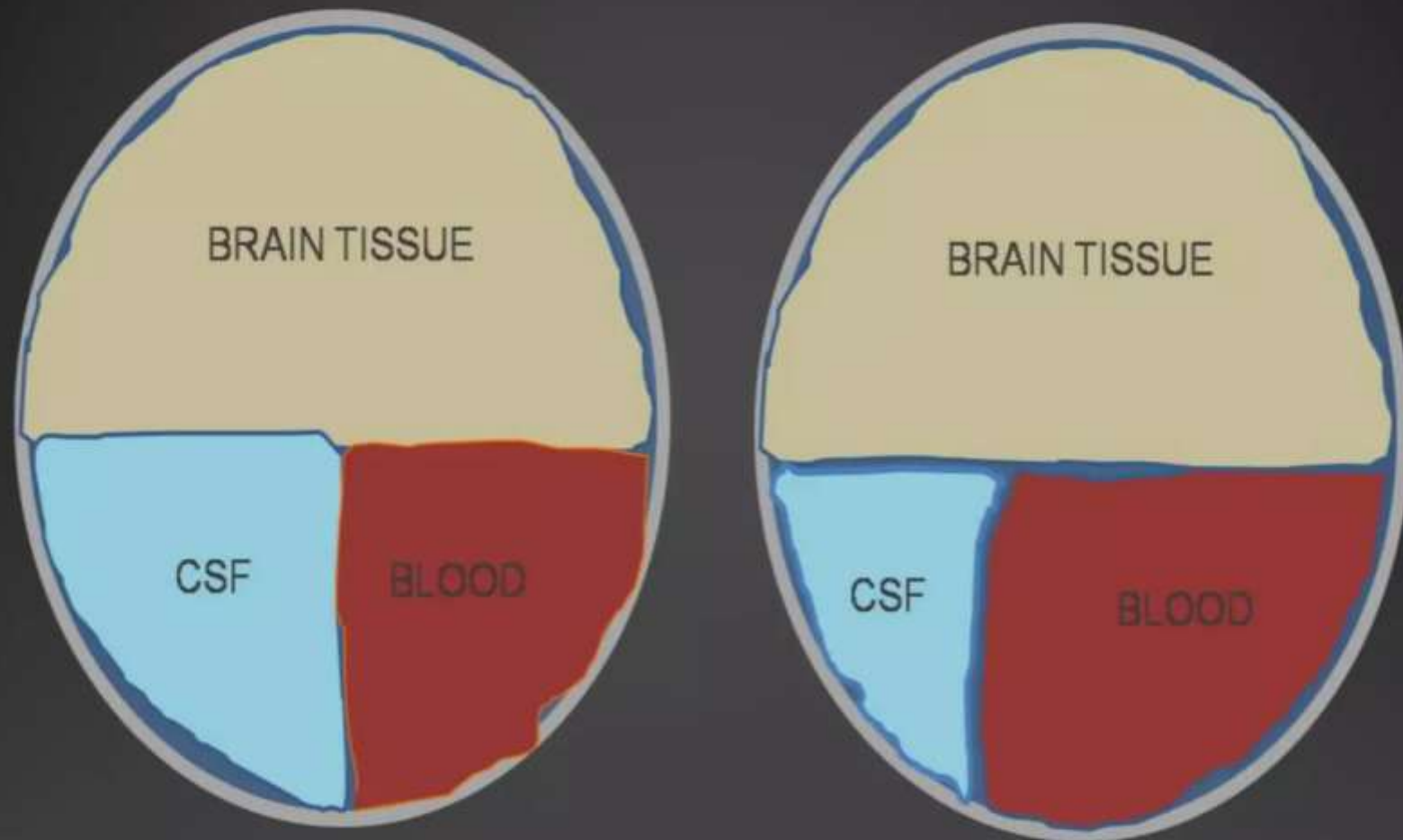
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- ▶ Monitor for pain and restlessness
 - ▶ Avoid the administration of morphine sulfate because it is respiratory depressant and may increase ICP.
 - ▶ Monitor for drainage from the nose or ears because this fluid may be CSF
 - ▶ Do not attempt to clean the nose, suction, or allow the client to blow the nose if drainage occurs
 - ▶ Do not clean the ear if drainage is noted, but apply a loose, dry sterile dressing
 - ▶ Check drainage for the presence of CSF
 - ▶ Notify the physician if drainage from ears or nose is noted
 - ▶ Instruct the client to avoid coughing because this increases ICP
 - ▶ Monitor for signs of infection
 - ▶ Prevent complications of immobility.



Table 60.1 Factors to assess in head injured patients

Clinical concern	Items to check	Steps to remedy
hypoxia or hypoventilation	ABG, respiratory rate	intubate any patient who has hypercarbia, hypoxemia, or is not localizing
hypotension or hypertension	BP, Hgb/Hct	transfuse patients with significant loss of blood volume
anemia	Hgb/Hct	transfuse patients with significant anemia
seizures	electrolytes, ASM levels	correct hyponatremia or hypoglycemia; administer ASMs when appropriate ^a
infection or hyperthermia	WBC, temperature	LP if meningitis is possible and no contraindications (p.1811)
spinal stability	spine X-rays	spine immobilization (spine board, cervical collar & sandbags...); patients with locked facets should be reduced if possible before transfer

THE MONRO-KELLIE HYPOTHESIS

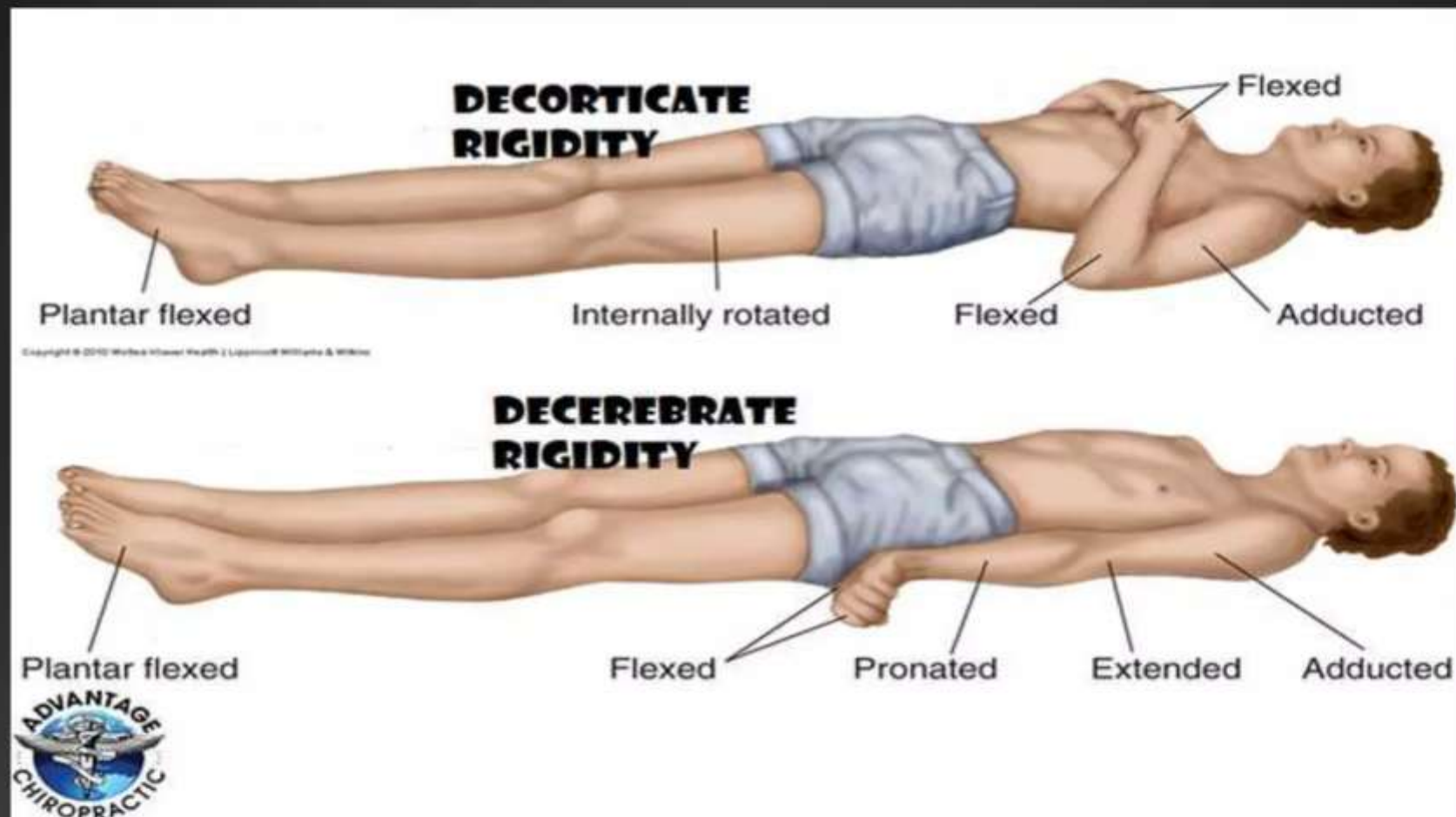


CLINICAL MANIFESTATIONS

- As neurologic function deteriorates further, the patient becomes comatose and exhibits abnormal motor responses in the form of **decortication**, **decerebration**, or flaccidity.
- When the coma is profound, with the pupils dilated and fixed and respirations impaired, death is usually inevitable.



DECORTICATION, DECEREBRATION POSTURE



OTHERS CLINICAL MANIFESTATIONS

1. Changes in LOC
2. Changes in vital signs
3. Pupillary changes (due to increased pressure on the optic and oculomotor nerves
4. Headache is increasing in intensity and aggravated by movements and straining.
5. Vomiting recurrent with little or nausea, esp. in early morning.



CLINICAL MANIFESTATIONS

6. Papilledema from optic nerve compression.
7. Restlessness, headache, forced breathing pattern, purposeless movements and mental cloudiness.
8. Seizures activity: focal or generalized
9. Decreased brain stem function (CN deficits such as loss of control reflexes and ability to swallow.
10. Pathologic reflexes: Babinski, grasp, chewing, sucking.



ASSESSMENT AND DIAGNOSTIC FINDINGS

1. The diagnostic studies used to determine the underlying cause of increased ICP .
2. Cerebral angiography
3. Computed tomography (CT) scanning
4. Magnetic resonance imaging (MRI) or positron emission tomography (PET).
5. Transcranial Doppler studies provide information about cerebral blood flow.



- Lumbar puncture is avoided in patients with increased ICP because the sudden release of pressure can cause the brain to herniate.



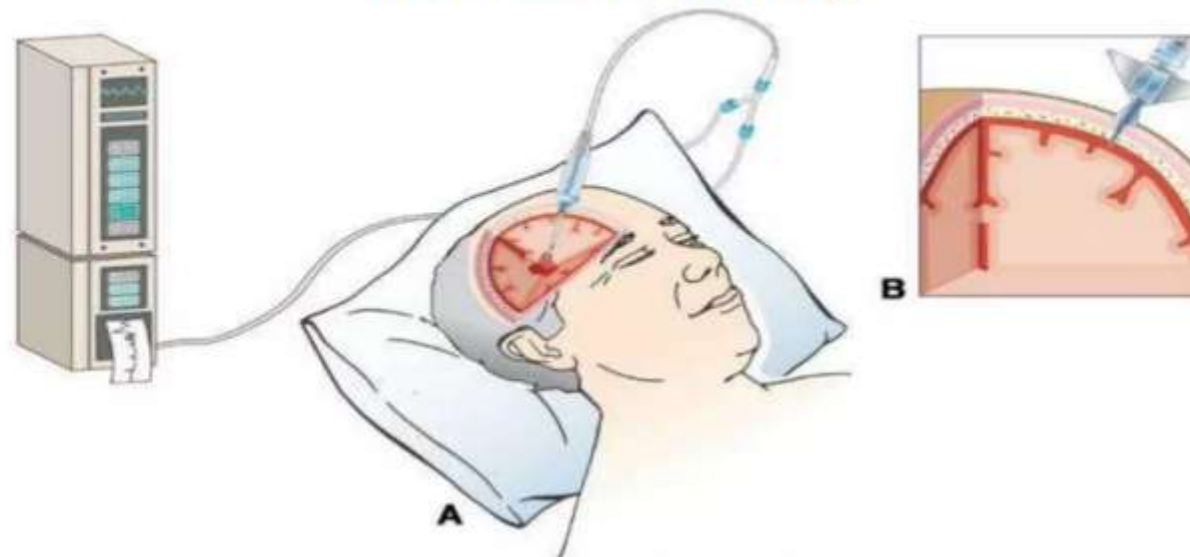
MONITORING ICP

- *An intraventricular catheter (ventriculostomy), a subarachnoid bolt, an epidural or subdural catheter, or a fiberoptic transducer-tipped catheter* placed in the subdural space or the ventricle can be used to monitor ICP.

ICP MONITORING

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ICP Monitoring



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ICP Management

- Prevention of secondary injury to brain becomes the primary goal of therapy following the treatment of any surgical lesions
- Cerebral Blood Flow 50cc/100g/min
- $CPP = MAP - ICP$
- Monro-Kellie doctrine

ICP Normal Values

- Infant <7.5 mm Hg
- Child <10 mm Hg
- Adult <15 mm Hg

Indications for ICP Monitoring

- Any pt with GCS < 8 and abnormal CT
- GCS < 8 with normal CT scan if age >45, motor posturing, or SBP < 90
- Treat ICP > 20-25 mmHg to maintain CPP > 70 mmHg

Treatment of ICP

■ Preemptive Measures

- ☐ HOB at 30 degrees
- ☐ PaCO₂ 35-40 mmHg
- ☐ Euvolemia
- ☐ Seizure prophylaxis
- ☐ Sedation, paralytics
- ☐ CPP > 70
- ☐ Prevent clot expansion with FFP, Novo VII, Vitamin K, Platelets as appropriate

Treatment of ICP

- CSF drainage from ventric. + hyperventilate to PaCO₂ 30-35 mmHg
- Mannitol 0.5-1 gm/kg to Ser Osm <320, Hypertonic saline
- Consider cooling the patient
- If present, remove surgical lesion, or consider hemicraniectomy
- Barbiturate coma with pentobarb
 - 10mg/kg/30 mins
 - 3 mg/kg q 1h x 3
 - 1 mg/kg/hr